Exp No: 8 Conduct Z test for the Data given using Python

AIM:

To conduct a one-sample Z-test in Python to test whether the average weight of a species of birds differs from the claimed population mean of 150 grams.

PROGRAM:

```
import numpy as np

import scipy.stats as stats

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151,
150, 149, 152, 151,

148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

population_mean = 150

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:

  print("Reject the null hypothesis: The average weight is significantly different
from 150 grams.")

else:
```

```
  print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

OUTPUT:

Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.

RESULT:

Thus a one-sample Z-test in Python to test whether the average weight of a species of birds differs from the claimed population mean of 150 grams is conducted.